# Supplementary Material for "Classical Learners, Convolutional Networks and Vision Transformers for ICROP-3 Staging of Retinopathy of Prematurity: An Audited Development across Four Public Cohorts"

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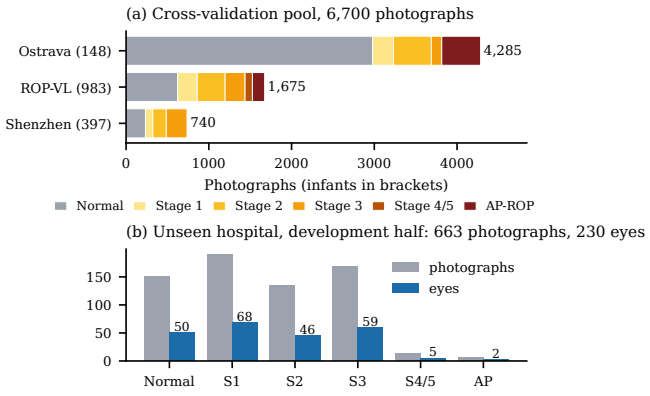



Fig. S1. Cross-validation pool by cohort and stage (a), and the held-out cohort's development half by stage (b): photographs and eyes.

## I. Contents

Tables S1–S7 and Figs. S3–S10 hold the material referred to from the main text; Table S8 maps the CLAIM 2024 items to where each is reported. All of it is generated by `scripts/paper4_figures.py` from the files named in the main text's Data and Code Availability statement.

## II. Data

Table S1 is the acquisition cascade of the main text's Fig. 1 in numbers; Fig. S1 shows the class composition of the cross-validation pool and of the held-out cohort's development half (the figure labels the latter "unseen hospital"); Fig. S2 places the generations in time.

## III. Generation 1: convolutional networks

Tables S2 and S3 are the June 2026 binary models that preceded six-class staging; both carry the caveats stated in their captions and in the main text's Table VIII.

## IV. Generation 2: vision transformers

Table S4 gives every arm's recipe and training cost; Figs. S3–S9 are the leave-one-source-out guard, the per-class recall of all arms, the nine confusion matrices, the ROC and precision–recall curves, the reliability diagrams, the training curves and the Grad-CAM strip referred to from the main text.

## V. Independent packages

Tables S5–S7 list every model of the three independent packages; the main text's Section IV-C states why none of them is comparable with the protocol results.

## VI. The served application

Fig. S10 is the results panel of the served application on a development-half photograph.

## VII. Reporting checklist

Table S8 maps the CLAIM 2024 items to where each is reported.

TABLE S1

ACQUISITION CASCADE IN NUMBERS, AND HOW THE ELIGIBLE PHOTOGRAPHS ARE USED. REMOVED PHOTOGRAPHS AT THE TRAINING-ELIGIBLE STEP STAY IN VALIDATION: EVERY PHOTOGRAPH OF THE POOL IS SCORED EXACTLY ONCE BY THE MODEL THAT NEVER TRAINED ON ITS INFANT.

| Step | Photos | Infants | Removed | Reason / note |
|---|---|---|---|---|
| Acquired | 13,037 | – | – | 11 publicly released sources |
| Stage-labelled cohorts | 10,427 | 2,015 | 2,610 | no ICROP stage label (plus-only, segmentation sets), too small to fold, re-encoded duplicates |
| Six-class eligible | 8,024 | 1,760 | 2,403 | other pathology 1,295, treated 1,063, stage 0 45 |
| Cross-validation pool | 6,700 | 1,528 | – | three cohorts; every photo in one validation fold |
| Training-eligible | 3,112 | 1,528 | 3,588 | per-infant caps: 2 normal, 20 per disease class; capped photos stay in validation |
| Held-out cohort, dev half | 663 | 116 | – | 230 eyes; checks and every reported external number |
| Held-out cohort, locked half | 661 | 116 | – | 230 eyes; opened once, Aug 2026, for the CNN generation |

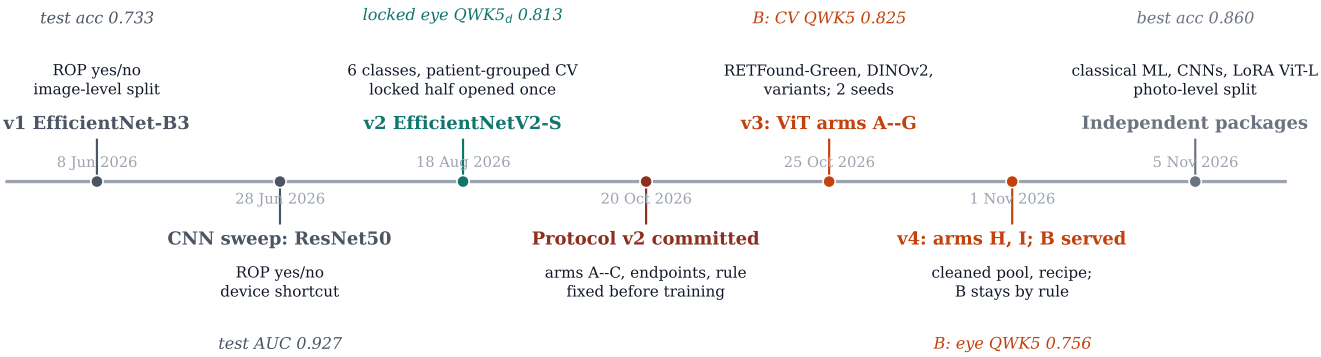



Fig. S2. The development in time, with each generation's split type and its headline under its own protocol. Numbers across generations are not comparable (main text, Table VIII).

TABLE S2

FIRST MODEL (JUNE 2026): EFFICIENTNET-B3, ONE SHARED MODEL FOR BINARY ROP AND DIABETIC RETINOPATHY (DR), IMAGE-LEVEL SPLIT. P / R: PRECISION / RECALL. RETIRED: IMAGE-LEVEL SPLIT AND TWO DISEASES IN ONE MODEL.

| Split | n | Acc. | mF1 | No ROP P / R | ROP P / R | ROP F1 |
|---|---|---|---|---|---|---|
| ROP val | 928 | 0.647 | 0.646 | 0.476 / 0.933 | 0.941 / 0.510 | 0.661 |
| ROP test | 1,502 | 0.733 | 0.731 | 0.644 / 0.913 | 0.891 / 0.585 | 0.706 |
| DR val, ep. 5 | – | 0.719 | 0.497 | – | – | – |

TABLE S3

JUNE 2026 BINARY CNN SWEEP (ROP YES/NO, OSTRAVA, SWEEP VALIDATION SPLIT; $QWK_2$: BINARY KAPPA) AND, *, THE DEPLOYED RESNET50 ON ITS OWN VALIDATION AND TEST SPLIT. INFLATED BY THE DEVICE SHORTCUT: IMAGE SIZE ALONE REACHED AUC 0.911 ON THE SAME SPLIT.

| Architecture | Acc. | mF1 | $QWK_2$ | AUC | Params (M) | CPU ms |
|---|---|---|---|---|---|---|
| efficientnet_b0 | 0.758 | 0.757 | 0.525 | 0.867 | 4.01 | 56.2 |
| efficientnet_v2_s | 0.706 | 0.701 | 0.431 | 0.870 | 20.18 | 130.1 |
| mobilenet_v3_large | 0.744 | 0.743 | 0.500 | 0.857 | 4.2 | 34.4 |
| resnet50 | 0.886 | 0.882 | 0.766 | 0.964 | 23.51 | 157.2 |
| densenet121 | 0.764 | 0.764 | 0.537 | 0.887 | 6.96 | 136 |
| resnet50* val | 0.747 | 0.740 | 0.499 | 0.842 | – | – |
| resnet50* test | 0.844 | 0.843 | 0.688 | 0.927 | – | – |

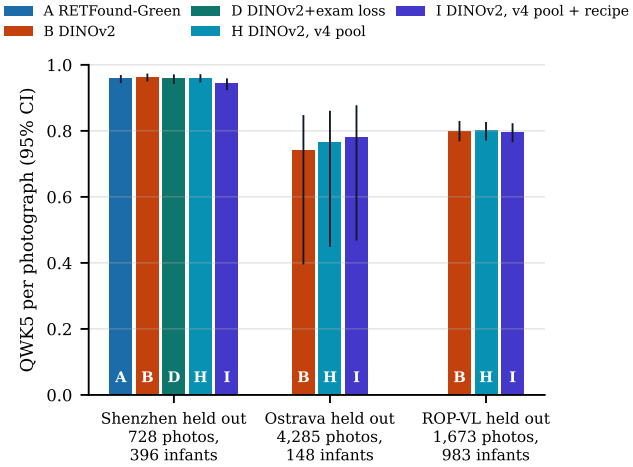



Fig. S3. Leave-one-source-out folds: each model trains on two cohorts and scores every photograph of the third (both seeds pooled). Stage 4/5 exists only in ROP-VL, so that fold scores it untrained.

TABLE S4

RECIPE AND COST OF THE TRANSFORMER ARMS. SHARED BY EVERY TRAINED ARM: CIRCLE-CROP, LETTERBOX, CLAHE; CLASS-BALANCED CROSS-ENTROPY; STRONG AND SITE-RANDOMIZING AUGMENTATION; BATCH 16; ADAMW; ONE-CYCLE COSINE SCHEDULE; LAST-EPOCH WEIGHTS. LLRD: LAYER-WISE LEARNING-RATE DECAY; LS: LABEL SMOOTHING. COST: MEAN OVER CROSS-VALIDATION FOLDS AND BOTH SEEDS ON KAGGLE TESLA T4 GPUS; C AND E WERE NOT TRAINED IN THIS GENERATION.

| Arm | Backbone | px | Norm. | Pool | Peak lr | LLRD | LS | Ep. | Differences from B's recipe | min/fold | img/s |
|---|---|---|---|---|---|---|---|---|---|---|---|
| A | `retfound_green` | 392 | half | v3 | 2e-4 | 0.8 | 0.05 | 20 | RETFound-Green weights | 32.1 | 38.6 |
| **B** | `dinov2_s` | 392 | imagenet | v3 | 1e-4 | 0.8 | 0.05 | 20 | reference recipe | 33.3 | 31.9 |
| C | `effnetv2s` | 384 | imagenet | v3 | 3e-4 | – | 0.05 | 20 | v2 flat softmax, natural sampler; saved predictions re-scored | – | – |
| D | `dinov2_s` | 392 | imagenet | v3 | 1e-4 | 0.8 | 0.05 | 20 | LSE bag loss on Ostrava sessions, $K$=8, $\tau$=1 | 32.5 | 29.0 |
| E | – | – | – | v3 | – | – | – | – | mean of A, B, C probabilities, matched by seed; no training | – | – |
| F† | `dinov2_s` | 392 | imagenet | v3 | 1e-4 | 0.8 | 0.05 | 20 | self-ensembling on unlabelled dev photos, EMA 0.99, conf. $\geq$0.9 | 61.7 | 14.2 |
| G | `dinov2_s` | 518 | imagenet | v3 | 1e-4 | 0.8 | 0.05 | 20 | 518 px | 58.1 | 19.2 |
| H | `dinov2_s` | 392 | imagenet | v4 | 1e-4 | 0.8 | 0.05 | 20 | v4 pool (per-device quality filter) | 34.1 | 31.2 |
| I | `dinov2_s` | 392 | imagenet | v4 | 1e-4 | 0.8 | 0.1 | 20 | EMA weights 0.999; 1 head-only epoch; label smoothing 0.1; rot. $\pm$180°, crop $\geq$0.85, gamma, noise, Fourier swap $p$ 0.5 | 34.8 | 32.1 |

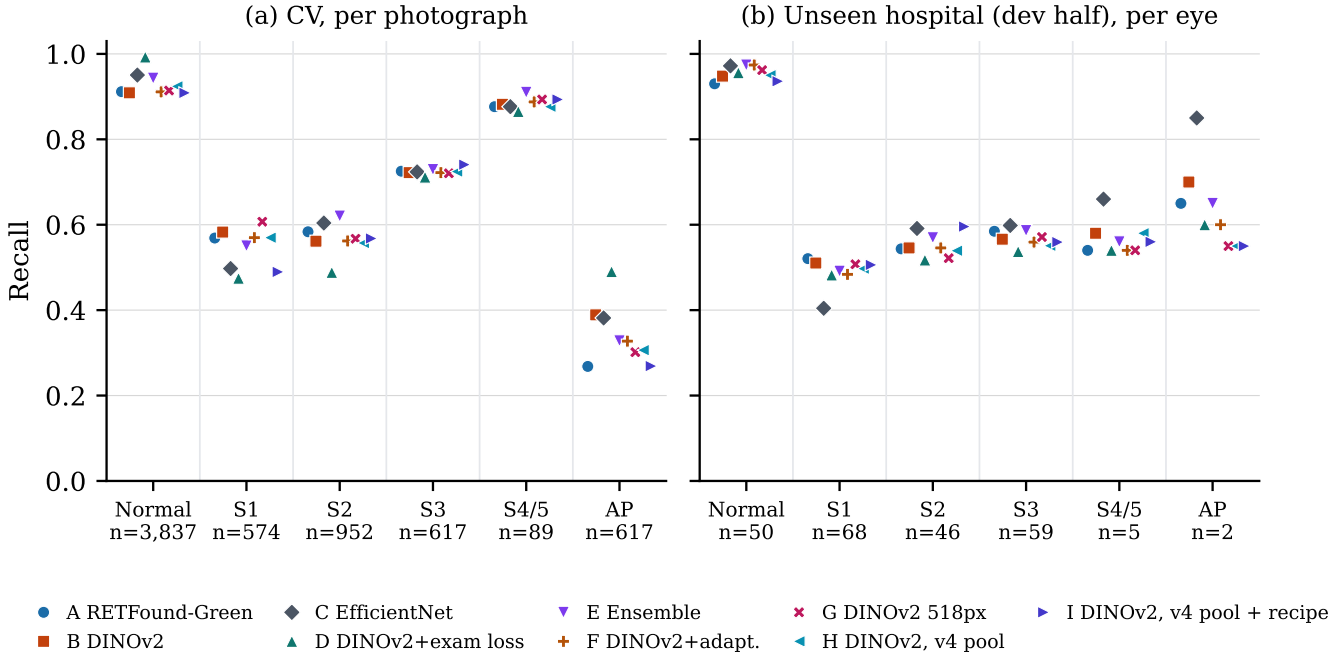



Fig. S4. Recall per class, all transformer arms: (a) cross-validation per photograph; (b) held-out cohort per eye. n below each class: photographs (a), eyes (b). Arm identity by color and marker; the values are in the main text's per-class table.

TABLE S5

CLASSICAL LEARNERS ON 428 HAND-CRAFTED FEATURES (INDEPENDENT PACKAGE; FOUR CLASSES; PHOTO-LEVEL 80/20 SPLIT; CV: FIVE-FOLD ACCURACY ON THE AUGMENTED TRAINING ROWS, WHICH RANKED THE LEARNERS; S: TRAINING SECONDS). NOT COMPARABLE WITH THE PROTOCOL RESULTS (MAIN TEXT, SEC. IV-C).

| | CV | Test, 629 photographs | | | | | |
|---|---|---|---|---|---|---|---|
| Model | Acc. | Acc. | Bal. acc. | mF1 | QWK | AUC | s |
| SVM (RBF) | 0.562 | 0.585 | 0.576 | 0.576 | 0.547 | 0.812 | 23 |
| Soft Voting (top-3) | 0.571 | 0.585 | 0.571 | 0.569 | 0.573 | 0.810 | 593 |
| Extra Trees | 0.542 | 0.574 | 0.561 | 0.555 | 0.502 | 0.790 | 6 |
| XGBoost | 0.562 | 0.568 | 0.554 | 0.552 | 0.535 | 0.804 | 130 |
| HistGradBoost | 0.566 | 0.568 | 0.552 | 0.550 | 0.551 | 0.807 | 459 |
| LightGBM | 0.567 | 0.566 | 0.555 | 0.555 | 0.563 | 0.810 | 39 |
| MLP | 0.534 | 0.558 | 0.547 | 0.549 | 0.531 | 0.795 | 13 |
| Stacking (top-3) | 0.567 | 0.556 | 0.543 | 0.541 | 0.556 | 0.807 | 2,107 |
| Random Forest | 0.540 | 0.555 | 0.540 | 0.532 | 0.500 | 0.786 | 9 |
| CatBoost | 0.547 | 0.522 | 0.511 | 0.508 | 0.458 | 0.784 | 33 |
| KNN | 0.518 | 0.501 | 0.493 | 0.489 | 0.403 | 0.736 | 5 |
| Logistic Reg. | 0.520 | 0.488 | 0.477 | 0.475 | 0.410 | 0.762 | 6 |

TABLE S6

INDEPENDENT CNN PACKAGE: IMAGENET-PRETRAINED CNNS (TIMM), 224 PX, UP TO 40 EPOCHS WITH EARLY STOPPING ON VALIDATION MACRO-F1; THE SOFT VOTE AVERAGES THE THREE WITH THE HIGHEST VALIDATION MACRO-F1. FOUR CLASSES; PHOTO-LEVEL SPLIT.

| | | | Val | Test, 315 photographs | | | | |
|---|---|---|---|---|---|---|---|---|
| Model | Par. (M) | Ep. | mF1 | Acc. | Bal. | mF1 | QWK | AUC |
| resnest50d | 25.4 | 32 | 0.821 | 0.835 | 0.827 | 0.828 | 0.898 | 0.956 |
| densenet121 | 7.0 | 36 | 0.822 | 0.822 | 0.816 | 0.815 | 0.880 | 0.942 |
| efficientnet_b2 | 7.7 | 40 | 0.832 | 0.819 | 0.812 | 0.811 | 0.884 | 0.943 |
| resnet50 | 23.5 | 40 | 0.827 | 0.810 | 0.802 | 0.800 | 0.860 | 0.951 |
| mobilenetv3_large_100 | 4.2 | 22 | 0.769 | 0.775 | 0.775 | 0.769 | 0.840 | 0.925 |
| convnext_tiny | 27.8 | 13 | 0.491 | 0.530 | 0.546 | 0.520 | 0.497 | 0.821 |
| Soft vote, top 3 | – | – | 0.856 | 0.825 | 0.821 | 0.818 | 0.881 | 0.959 |

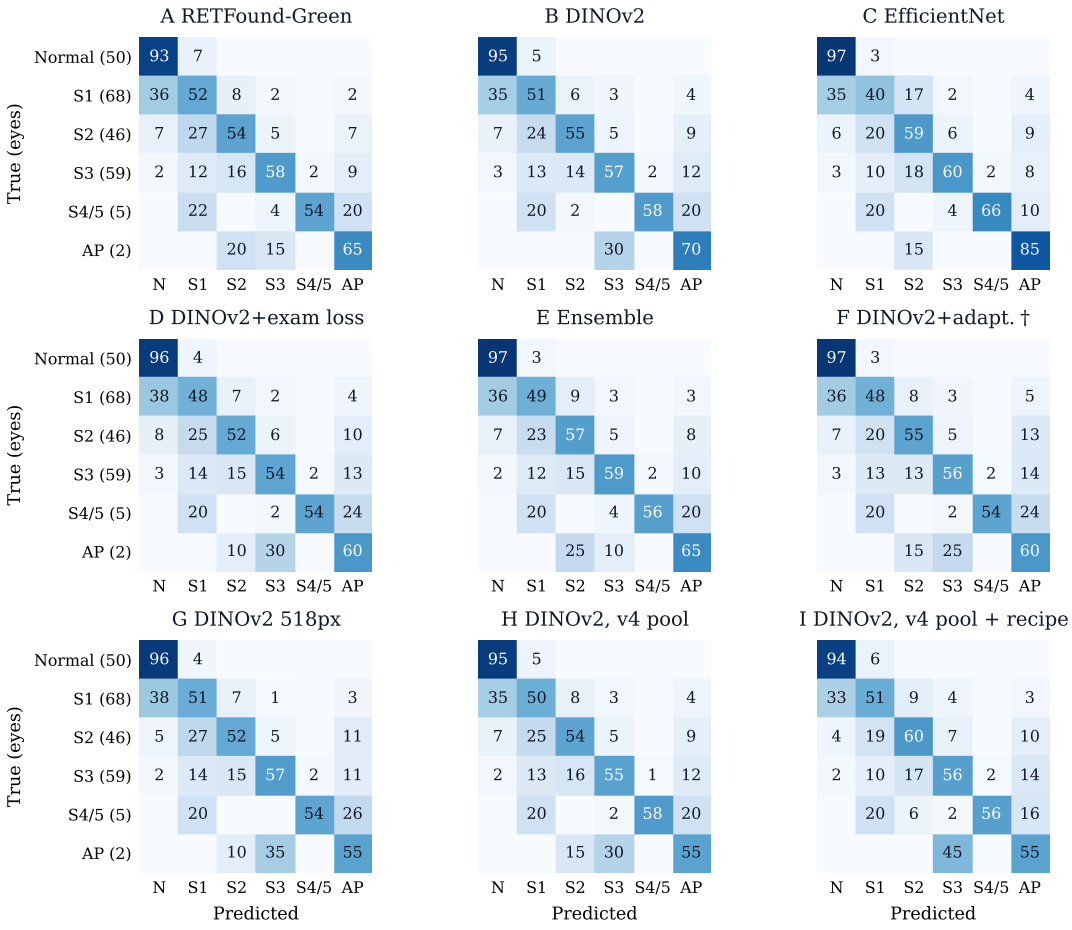



Fig. S5. Eye-level confusion matrices at the held-out cohort (development half), all nine arms. Cell: share of the true-class row in %, averaged over each arm's 10 models (diagonal = recall); blank = 0. True-class eye counts in parentheses. †Transductive.

TABLE S7

INDEPENDENT FOUNDATION-MODEL PACKAGE: RETFOUND-DINOV2 VIT-L/14 FROZEN, LORA ADAPTERS (RANK 16, ALPHA 32 UNLESS STATED) ON ATTENTION AND MLP LAYERS, LINEAR FOUR-CLASS HEAD; BEST EPOCH BY VALIDATION MACRO-F1; FOUR-VIEW FLIP TTA. PROBES: 10 EPOCHS, ONE CHANGE EACH. FOUR CLASSES; PHOTO-LEVEL SPLIT.

| | | Trainable | | | Val | Test, 315 photographs | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Run | Change | (M) | px | Epochs | mF1 | Acc. | Bal. acc. | mF1 | QWK | AUC |
| *Stage-2 probes, 10 epochs, seed 42 (one change each vs the first row)* | | | | | | | | | | |
| p mlp | LoRA on attention + MLP (probe base) | 6.30 | 224 | 10 | 0.810 | 0.797 | 0.798 | 0.792 | 0.862 | 0.947 |
| p mlp r32 | rank 32, alpha 64 | 12.59 | 224 | 10 | 0.819 | 0.816 | 0.811 | 0.810 | 0.867 | 0.950 |
| p mlp dp1 | drop-path 0.1 | 6.30 | 224 | 10 | 0.830 | 0.797 | 0.799 | 0.793 | 0.861 | 0.951 |
| p mlp lr6e4 | lr 6e-4 | 6.30 | 224 | 10 | 0.829 | 0.829 | 0.825 | 0.823 | 0.897 | 0.958 |
| p mlp img280 | 280 px input | 6.30 | 280 | 10 | 0.834 | 0.822 | 0.821 | 0.818 | 0.875 | 0.959 |
| p mlp ord | ordinal auxiliary loss 0.3 | 6.30 | 224 | 10 | 0.807 | 0.806 | 0.807 | 0.803 | 0.876 | 0.951 |
| p mlp combo | r32 + dp 0.1 + lr 6e-4 + 280 px | 12.59 | 280 | 10 | 0.623 | 0.651 | 0.662 | 0.617 | 0.761 | 0.900 |
| *Full-length runs, 30 epochs max* | | | | | | | | | | |
| final base s42 | attention-only LoRA, 30 ep | 2.37 | 224 | 30 | 0.817 | 0.825 | 0.818 | 0.818 | 0.900 | 0.956 |
| final mlp s42 | attention + MLP, 224 px, 30 ep | 6.30 | 224 | 25 | 0.816 | 0.778 | 0.763 | 0.765 | 0.860 | 0.948 |
| final mlp img280 s42 | chosen config, seed 42 | 6.30 | 280 | 25 | 0.846 | 0.787 | 0.787 | 0.779 | 0.870 | 0.950 |
| final mlp img280 s1 | chosen config, seed 1 | 6.30 | 280 | 30 | 0.841 | 0.822 | 0.818 | 0.815 | 0.888 | 0.952 |
| final mlp img280 s2 | chosen config, seed 2 | 6.30 | 280 | 30 | 0.846 | 0.832 | 0.827 | 0.826 | 0.889 | 0.951 |
| 3 seeds averaged | chosen config, seeds 42, 1, 2 | – | – | – | 0.839 | 0.844 | 0.842 | 0.839 | 0.900 | 0.964 |
| 3 seeds + 4 CNNs | plus EffNet-B2, ResNet50, DenseNet121, ResNeSt-50d | – | – | – | 0.863 | 0.860 | 0.859 | 0.855 | 0.908 | 0.972 |

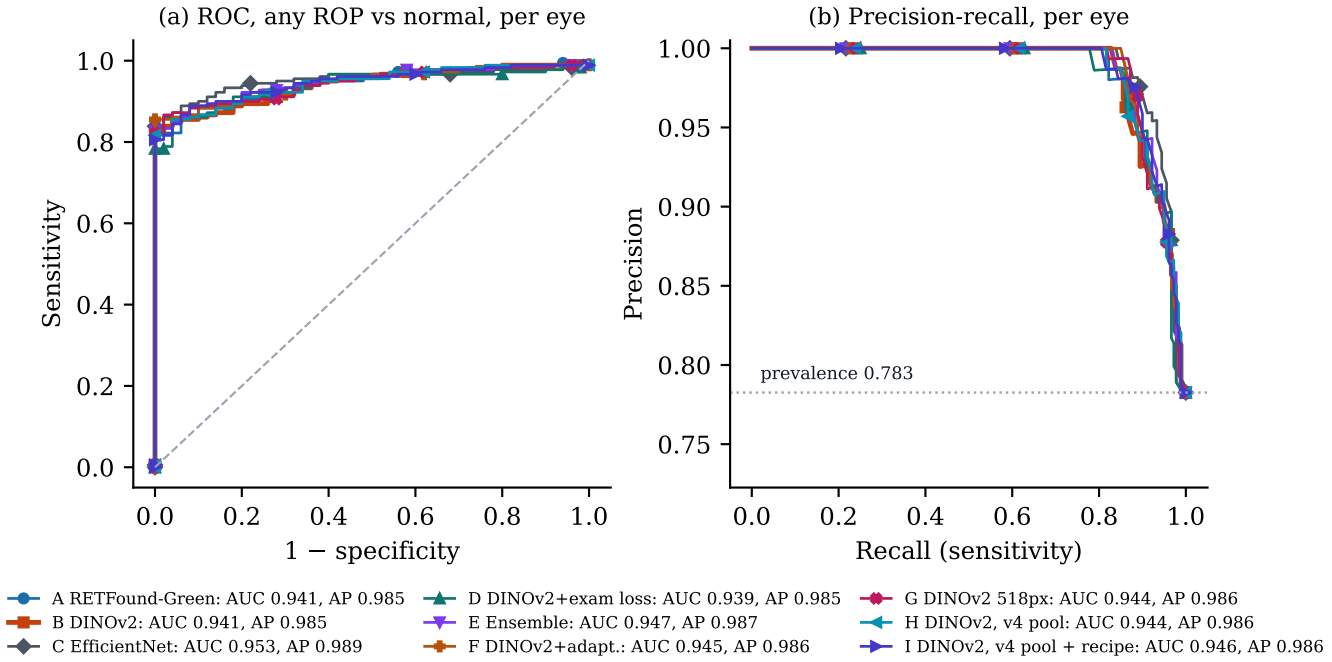



Fig. S6. Any ROP vs normal per eye at the held-out cohort (development half), all arms: ROC and precision–recall, eye score averaged over each arm's 10 models. Dotted: prevalence. Average precision is driven by the 78% prevalence and is not informative here.

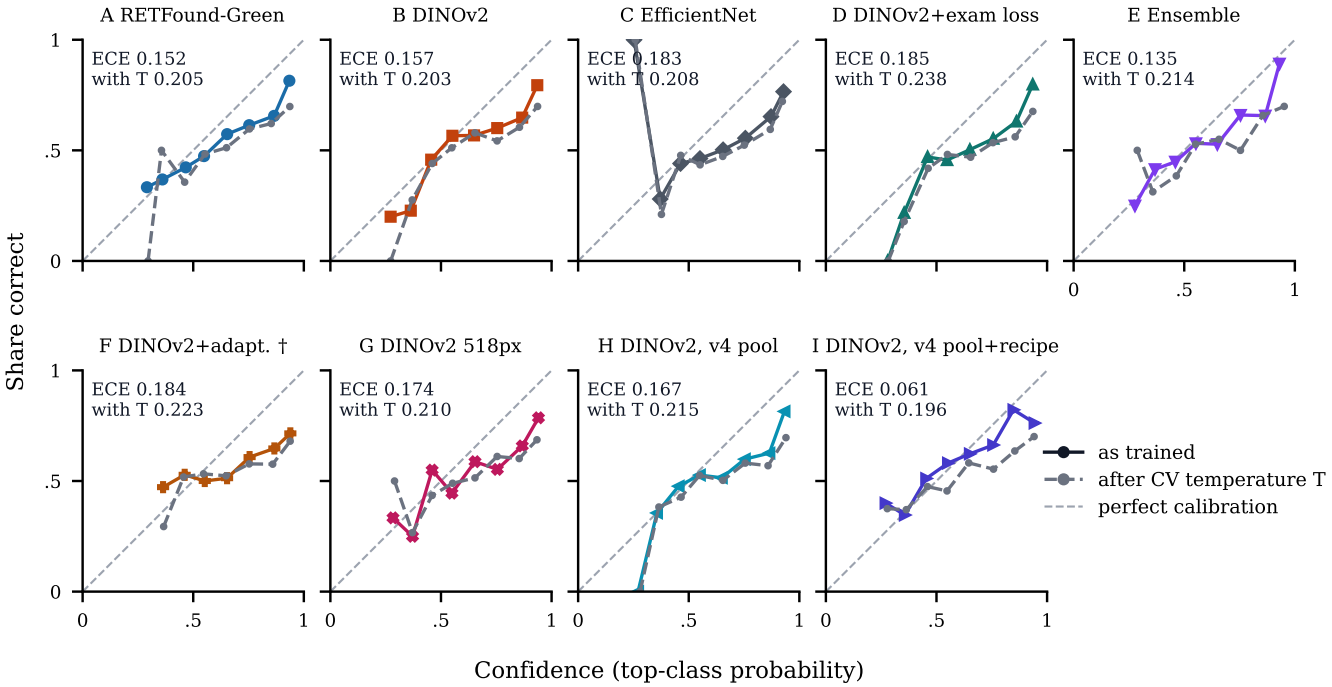



Fig. S7. Reliability of the eye-level top-class probability at the held-out cohort, ten equal-width bins, seeds pooled: as trained (color) and after each seed's cross-validation temperature (gray). The temperature makes every arm more over-confident at the new site.

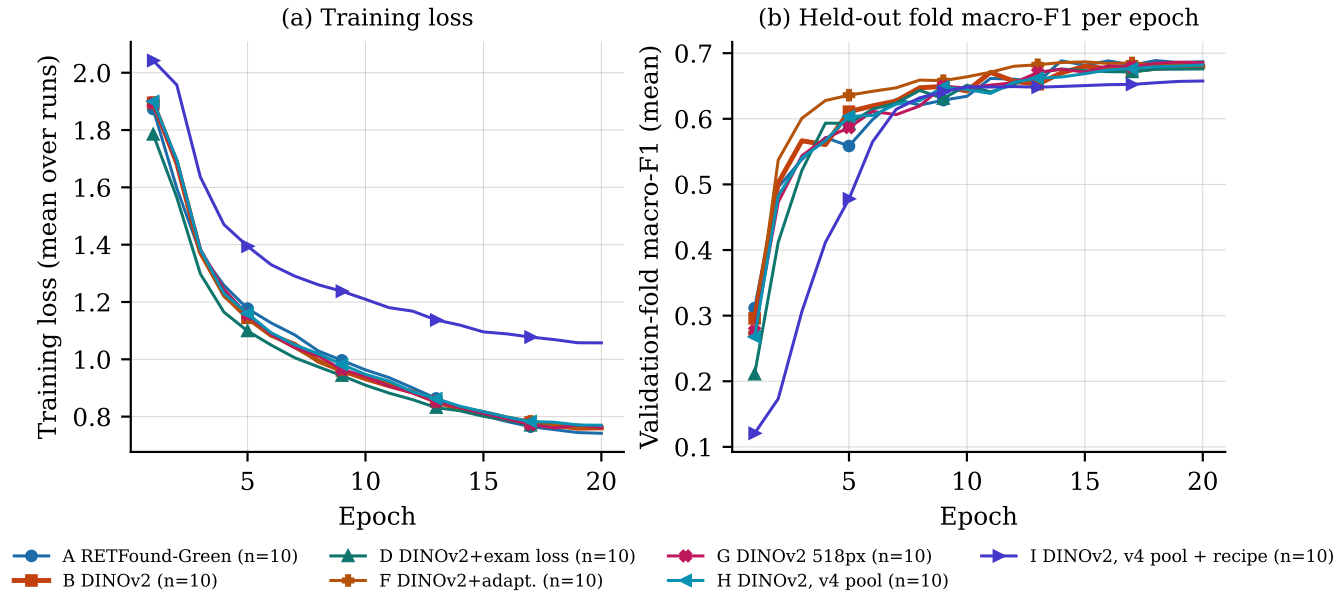



Fig. S8. Training curves, mean over each arm's 10 runs (five folds × two seeds): (a) training loss; (b) macro-F1 on the held-out fold per epoch.

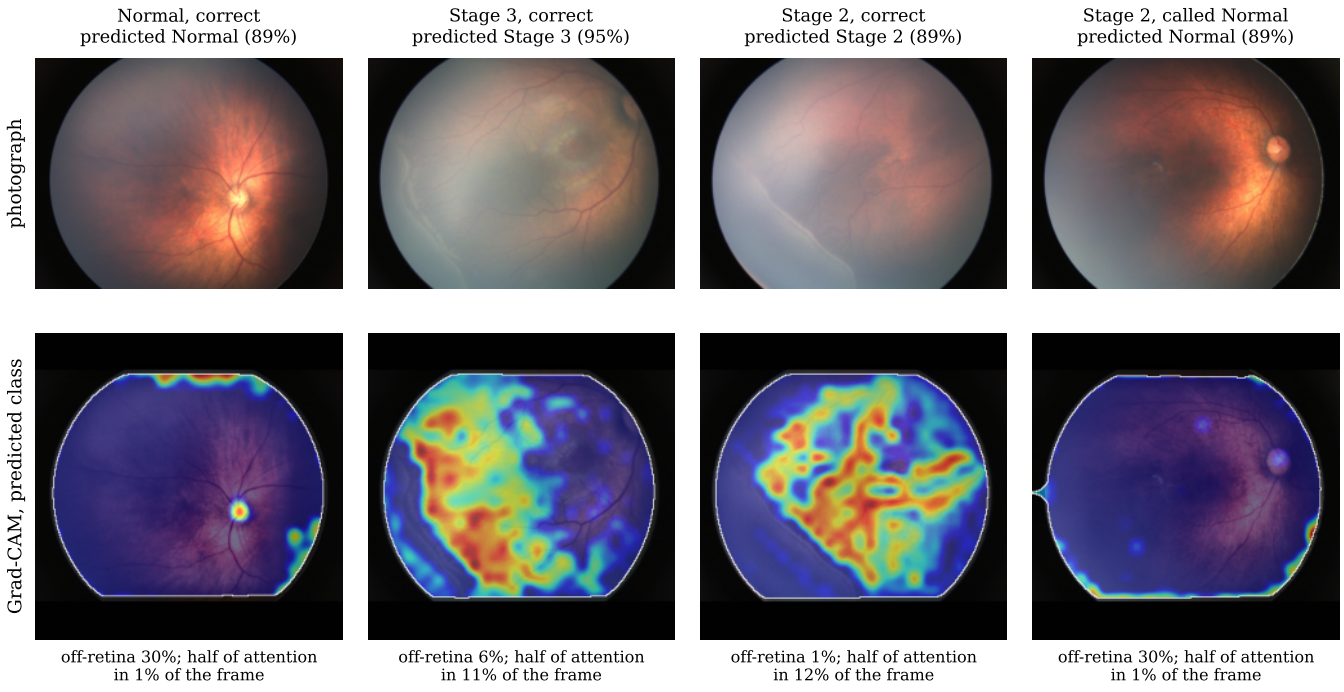



Fig. S9. Grad-CAM of the served ensemble (arm B, seed 42, five checkpoints) for development-half photographs, chosen by a fixed rule (the most confident example of each outcome: a correct normal, a correct stage 3, a correct stage 2, and a stage-2 photograph called normal). Maps are colored inside the retinal field only; under each, the share of attention outside the retina and how much of the frame holds half of it. Maps show what moved the score; they are not evidence of correctness and were not validated against ridge annotations.

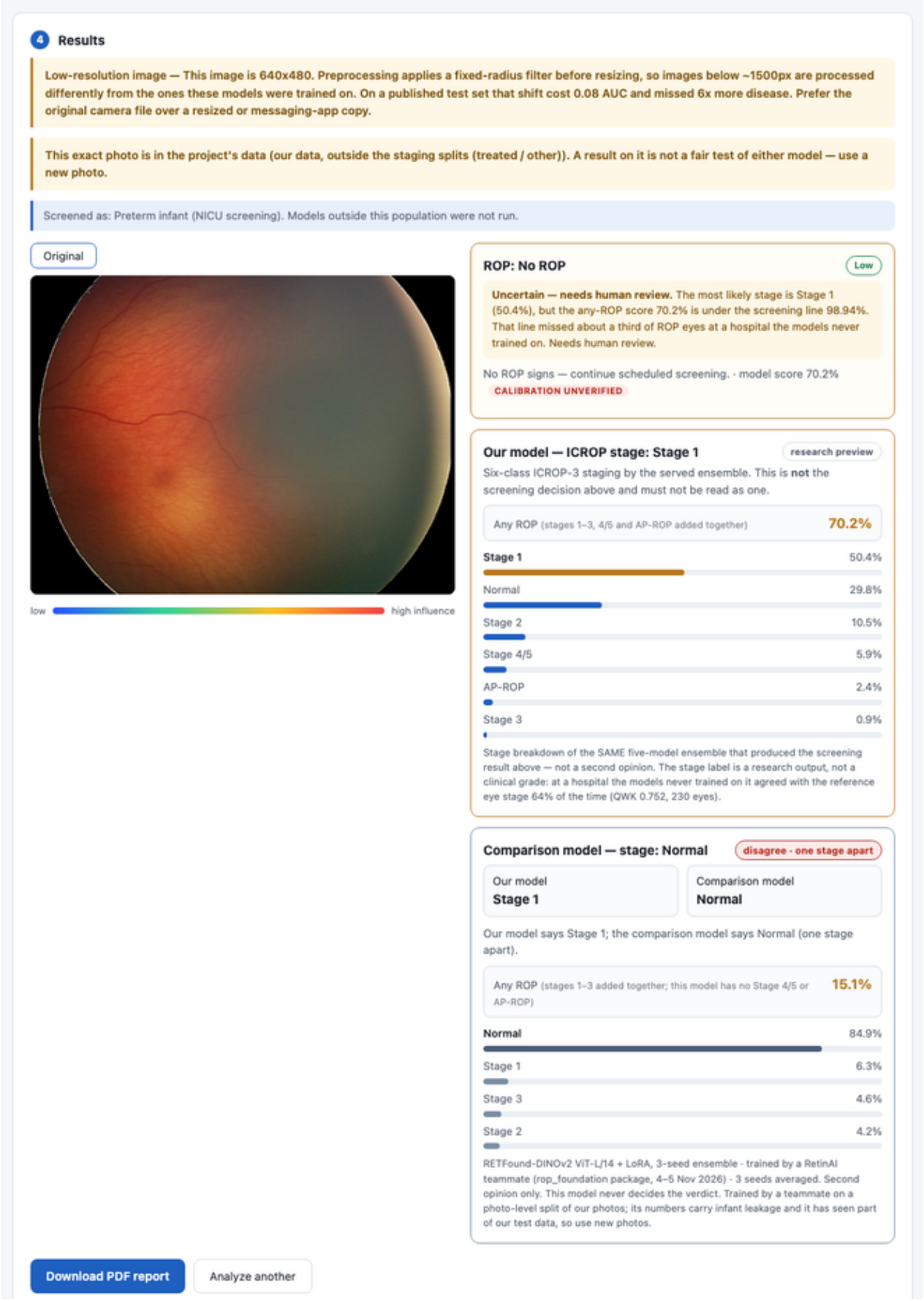



Fig. S10. The served application on a development-half photograph: verdict against the fixed operating point with the measured sensitivities, the served model's stage, the comparison model's stage with the agreement mark, and the seen-photo notice. Illustrative screenshot; the panel text is not meant to be read at print size. Research prototype; not a medical device.

TABLE S8

CLAIM 2024 ITEMS RELEVANT TO THIS STUDY AND WHERE EACH IS REPORTED (MAIN TEXT UNLESS STATED). ITEMS NOT APPLICABLE TO A RETROSPECTIVE STUDY OF PUBLIC DATA ARE OMITTED.

| Item | Where |
|---|---|
| Study design, data sources, eligibility | Sec. III-A (cascade, Fig. 1; Table S1 here), Table I |
| Reference standard, graders, label unit | Sec. III-A and III-B: labels as released; grader counts not stated in the releases; exclusions from released diagnosis codes, not visual review |
| Demographics | Not available in the releases except ROP-VL (gestational age, birth weight), not used; stated in Sec. III-A |
| Data partitions, leakage control | Sec. III-C (patient-grouped folds, held-out cohort, duplicate exclusion), Sec. III-D (audit, Table II) |
| Preprocessing, augmentation | Sec. III-E, Fig. 2 |
| Model architectures and training | Sec. IV; Table S4 here |
| Model selection rule, pre-specification | Sec. IV-B (protocol commit and amendments, selection rule, margins) |
| Evaluation metrics, intervals, multiplicity | Sec. IV-B and V-C (bootstrap over infants, paired contrasts, minimum detectable difference, Holm policy) |
| External evaluation | Sec. V-C–V-E (held-out cohort, development half; locked half unscored for the transformers) |
| Calibration | Sec. V-E; Fig. S7 here |
| Failure analysis | Sec. V-D (confusion by stage); Fig. S9 here (qualitative, not validated) |
| Limitations | Sec. VI |
| Data, code, model availability | Data and Code Availability (repository, generator scripts, weight hashes) |
| AI-use disclosure | Acknowledgments |